Byungkook Oh

Assistant Professor
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"Unifying Graph & Language Intelligence toward knowledge-driven AI, enabling human-like reasoning for the future of AGI."

Research Interests

- Knowledge Representation & Neuro-Symbolic AI
- Knowledge-enhanced NLP Applications
- Information Retrieval & Recommendation

Education & Experience

Konkuk University, Seoul, Republic of Korea — *Mar. 2024 - Current*
Assistant Professor

Samsung Research, Seoul, Republic of Korea — *Jan. 2021 - Jan. 2024*
AI Research Scientist (Staff Engineer)

Yonsei University, Seoul, Republic of Korea — *Oct. 2020 - Dec. 2020*
Postdoctoral Researcher

Yonsei University, Seoul, Republic of Korea — *Mar. 2015 - Aug. 2020*
Ph.D. in Computer Science, GPA: 4.15/4.3
- Dissertation: Context-Aware Relational Learning for Knowledge Graph Completion
- Advisor: Prof. Kyong-Ho Lee

Yonsei University, Seoul, Republic of Korea — *Mar. 2011 - Feb. 2015*
B.S. in Computer Science, Major GPA: 3.72/4.3

Publications (International)

[35] Haeyoon Koo, Donghyun Lee, Bonyou Koo, Hye-Yoon Baek, Dongcheon Lee, and **Byungkook Oh***, "**Information-Adaptive Textual Tokenization for Generative Recommendation under Text Sparsity**." *(Under Review)*

[34] Hye-Yoon Baek, Sangjun Ji, Jimyeung Seo, Haeyoon Koo, Xiongnan Jin, and **Byungkook Oh***, "**Internalizing Negation-Gated Logical Rules into LLMs for Document-Level Relation Extraction**." *(Under Review)*

[33] Jinho Choi, Hye-Yoon Baek, Sang-Hyun Park, Sangjune Kim, Hyeongjun Yang, and **Byungkook Oh***, "**Causality-Guided Spurious Evidence Pruning with GFlowNet for Faithful Graph Retrieval**." *(Under Review)*

[32] Bonyou Koo, Jaejin Lee, Hae-Yoon Koo, Dongcheon Lee, Jimyeung Seo, and **Byungkook Oh***, "**An Empirical Analysis of Stability and Sensitivity in Node-level Graph Anomaly Detection**." *(Under Review)*

[31] Wonseok Son‡, Haeyoon Koo‡, Jeonghyeon Park, Jungkyu Han, **Byungkook Oh**, and Sejin Chun*, "**Top-HiCL: Hierarchical Contrastive Learning for Job-Skill Matching with Topological Awareness**." *(Under Review)*

[30] Hyeongjun Yang, Donghyun Kim, Kyuhwan Yeom, Kyong-Ho Lee*, **Byungkook Oh**, and Xiongnan Jin "**Reliable Distillation of Recommendation Rationales from Large Language Models for Conversational Recommendation**." IEEE Transactions on Knowledge and Data Engineering, 2026 *(Under Revision)*

[29] Sangjun Ji, Sangjune Kim, Youngho Lee, Bonyou Koo, Xiongnan Jin, and **Byungkook Oh***, "**Joint Global-Local Representations via Relation-Entity Pair Encoding for Hyper-Relational KGs**." ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2026 — Research Track (Acceptance Rate: 18.5%)

[28] Dong-Cheon Lee, Ji-Yeon Park, Hye-Yoon Baek, Jimyeung Seo, Seyeong Kim, and **Byungkook Oh***, "**Diversifying Differentiable Graph Retrieval with Topic-Adaptive Multi-Intent Learning**." The ACM Web Conference (WWW), 2026 — Research Track (Acceptance Rate: 20.1% = 676/3,370)

[27] Jimyeung Seo, Eun-Yeong Jo, Hye-Yoon Baek, Dongcheon Lee, Xiongnan Jin, and **Byungkook Oh***, "**Image-Guided Debiasing Distillation with Preference Alignment Across Multi-News Histories**." ACM International Conference on Web Search and Data Mining (WSDM), 2026 — Full Paper (Acceptance Rate: 16.3% = 100/613)

[26] Wooseok Jeong‡, Young-Jin Kim‡, Hae-Yoon Koo, Jimyeung Seo, Jinho Choi, and **Byungkook Oh***, "**Interaction-Grounded Semantic Graph Refinement for LLM-based Recommendation**." IEEE Access, 2025

[25] Hye-Yoon Baek, Jinho Choi, Jimyeung Seo, Xiongnan Jin, Dongcheon Lee, and **Byungkook Oh***, "**Relation-Faceted Graph Pooling with LLM Guidance for Dynamic Span-Aware Information Extraction**." ACM International Conference on Information and Knowledge Management (CIKM), 2025 — Full Research Paper (Acceptance Rate: 27.2% = 443/1,627)

[24] Xiongnan Jin, Zhilin Wang, Jinpeng Chen, Liu Yang, Byungkook Oh, Seung-won Hwang, and **Jianqiang Li***, "**HLMEA: Unsupervised Entity Alignment Based on Hybrid Language Models**." AAAI Conference on Artificial Intelligence (AAAI), 2025 — Main Technical Track (Acceptance Rate: 23.4% = 3,032/12,957)

[23] Xiongnan Jin, Zhilin Wang, Manni Duan, Yan Shao, Xingyun Hong, Yongheng Wang*, and **Byungkook Oh**, "**A survey on knowledge graph evolution proliferation, dynamic embedding, and versioning**." nternational Journal of Web and Grid Services (IJWGS), 2025

[22] Gayeon Park, Hyeongjun Yang, Kyuhwan Yeom, Myeongheon Jeon, Yunjeong Ko, **Byungkook Oh**, and Kyong-Ho Lee*, "**Knowledge-constrained Interest-aware Multi-Behavior Recommendation with Behavior Pattern Identification**." Information Sciences, 2025

[21] Hyeongjun Yang, Yerim Lee, Gayeon Park, TaeYoung Kim, Heesun Kim, Kyong-Ho Lee*, and **Byungkook Oh**, "**Granular Intents Learning via Mutual Information Maximization for Knowledge-aware Recommendation**." Knowledge-Based Systems, 2024

[20] Kyuhwan Yeom, Hyeongjun Yang, Gayeon Park, Myeongheon Jeon, Yunjeong Ko, **Byungkook Oh**, and Kyong-Ho Lee*, "**Embedding Two-View Knowledge Graphs with Class Inheritance and Structural Similarity**." ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2024 — Research Track (Acceptance Rate: 20.1% = 411/2,046)

[19] Heesoo Won, **Byungkook Oh**, Hyeongjun Yang, and Kyong-Ho Lee*, "**Cross-modal contrastive learning for aspect-based recommendation**." Information Fusion, 99, 2023.11 *(IF: 18.6)*

[18] **Byungkook Oh***, Yul Kim, and Bumki Min, "**Dual-Oriented Contrast for Recommendation with A Stop-Gradient Operation**." In CIKM 2023 *(full)*

[17] Seungmin Seo, **Byungkook Oh**, Jeongbeom Jeoung, Donghyun Kim, Kyong-Ho Lee*, Dong-Hoon Shin, and Yeonsoo Lee, "**Active learning for cross-sentence n-ary relation extraction**." Information Sciences, 645, 2023.10 *(IF: 8.1)*

[16] **Byungkook Oh***, Ilhyun Suh, Kihoon Cha, Junbeom Kim, Goeon Park, and Sihyun Jeong, "**Pre-training local and non-local geographical influences with contrastive learning**." Knowledge-Based Systems, 259, 2023.01 *(IF: 8.8)*

[15] Seungmin Seo, **Byungkook Oh**, Eunju Jo, Sanghak Lee; Dongho Lee; Kyong-Ho Lee; Donghoon Shin, and Yeonsoo Lee, "**Active Learning for Knowledge Graph Schema Expansion**." IEEE Transactions on Knowledge and Data Engineering, 34(12), 2022.12 *(IF: 8.9)*

[14] Byungkook Oh, Seungmin Seo, Jimin Hwang, Dongho Lee, and Kyong-Ho Lee, "**Open-world knowledge graph completion for unseen entities and relations via attentive feature aggregation**." Information Sciences, 586, 2022.03 *(IF: 8.1)*

[13] Junbeom Kim, Sihyun Jeong, Goeon Park, Kihoon Cha, Ilhyun Suh, and **Byungkook Oh**, "**DynaPosGNN: Dynamic-Positional GNN for Next POI Recommendation**." In ICDM Workshops 2021 *(eight pages)*

[12] Sungkwang Eom, **Byungkook Oh**, Sangjin Shin, and Kyong-Ho Lee, "**Multi-task learning for spatial events prediction from social data**." Information Sciences, 581, 2021.12 *(IF: 8.1)*

[11] **Byungkook Oh**, Jimin Hwang, Seungmin Seo, Sejin Chun, and Kyong-Ho Lee, "**Inductive Gaussian representation of user-specific information for personalized stress-level prediction**." Expert Systems with Applications, 178, 2021.09 *(IF: 8.5)*

[10] Dongho Lee, **Byungkook Oh**, Seungmin Seo, and Kyong-Ho Lee, "**News Recommendation with Topic-Enriched Knowledge Graphs**." In CIKM 2020 *(full)*

[9] Sanghak Lee, Seungmin Seo, **Byungkook Oh**, Kyong-Ho Lee, Donghoon Shin, and Yeonsoo Lee, "**Cross-sentence N-ary Relation Extraction using Entity Link and Discourse Relation**." In CIKM 2020 *(full)*

[8] Seungmin Seo, **Byungkook Oh**, and Kyong-Ho Lee, "**Reliable Knowledge Graph Path Representation Learning**." IEEE Access, 8, 2020.02 *(IF: 3.9)*

[7] Xiongnan Jin, **Byungkook Oh**, Sanghak Lee, Dongho Lee, Kyong-Ho Lee, and Liang Chen, "**Learning Region Similarity over Spatial Knowledge Graphs with Hierarchical Types and Semantic Relations**." In CIKM 2019 *(full)*

[6] **Byungkook Oh**, Seungmin Seo, Cheolheon Shin, Eunju Jo, and Kyong-Ho Lee, "**Topic-Guided Coherence Modeling for Sentence Ordering by Preserving Global and Local Information**." In EMNLP 2019 *(oral)*

[5] **Byungkook Oh**, Sangjin Shin, Sungkwang Eom, Jooik Jung, Minjae Song, Seungmin Seo, and Kyong-Ho Lee, "**Reliable TF-based recommender system for capturing complex correlations among contexts**." Journal of Intelligent Information Systems, 52(2), 2019.04 *(IF: 3.4)*

[4] **Byungkook Oh**, Seungmin Seo, and Kyong-Ho Lee, "**Knowledge Graph Completion by Context-Aware Convolutional Learning with Multi-Hop Neighborhoods**." In CIKM 2018 *(full)*

[3] Minjung Kim, **Byungkook Oh**, Jooik Jung, and Kyong-Ho Lee, "**Outlier-robust web service selection based on a probabilistic QoS model**." International Journal of Web and Grid Services, 12(2), 2016.05 *(IF: 1.0)*

[2] Seungmin Seo, Sejin Chun, **Byungkook Oh**, and Kyong-Ho Lee "**SDPA: Sensor Data Processing Architecture for Modeling Semantic Data from Sensor Steams**." In IRI 2015

[1] Sejin Chun, Seungmin Seo, **Byungkook Oh**, and Kyong-Ho Lee, "**Semantic description, discovery and integration for the Internet of Things**." In ICSC 2015

Publications (Domestic)

[7] Seyeong Kim, Dongcheon Lee, Jengwon hwang, Sengho kang, and **Byungkook Oh**, "**객체 탐지를 통한 PCB 부품 단위 이상 탐지 파이프라인 제안** *(Proposal of PCB component-level anomaly detection pipeline via object detection)*." In KCC 2026

[6] Yewon Sa, Hae-Yoon Koo, Insung Han, Jungsik Lee, Sung-Jin Park, Sun-Young Im, and **Byungkook Oh**, "**멀티모달 의미 확장 기반의 해석 가능한 계층적 Textual ID 생성 기법** *(Interpretable Hierarchical Textual ID Generation via Multimodal Semantic Enrichment)*." In KCC 2026

[5] Sangjun Ji, Hye-Yoon Baek, Donghyun Lee, and **Byungkook Oh**, "**문서 수준 관계 추출을 위한 멘션-문맥 하이퍼그래프 기반 집계 기법** *(Mention-Context Hypergraph Aggregation for Document-level Relation Extraction)*." In KCC 2026

[4] Yeeun Kim, Bonyou Koo, Insung Han, Jungsik Lee, Sung-Jin Park, Sun-Young Im, and **Byungkook Oh**, "**의미·빈도 정보를 반영한 로그 복합 임베딩 기반 이상 탐지 기법** *(Semantic and Frequency-Aware Hybrid Log Embedding for Anomaly Detection)*." In KCC 2026

[3] Jaejin Lee, Bonyou Koo, Insung Han, Jungsik Lee, Sung-Jin Park, Sun-Young Im, and **Byungkook Oh**, "**언어 모델과 그래프 신경망의 태스크 정렬을 통한 프로비넌스 그래프 APT 탐지** *(Provenance Graph-based APT Detection via Task Alignment of Language Models and Graph Neural Networks)*." In KCC 2026

[2] Donghyun Lee, Hae-Yoon Koo, and **Byungkook Oh**, "**그래프 패턴 머신 기반 서브구조 복원을 이용한 그래프 노드 이상 탐지** *(Graph Pattern Machine-based Node Anomaly Detection via Substructure Reconstruction)*." In KCC 2026

[1] Minjung Kim, Xiongnan Jin, Jooik Jung, **Byungkook Oh**, and Kyong-Ho Lee, "**확률적 QoS 모델 기반 웹서비스 선택 방법 (Web Service Selection based on a Probabilistic QoS Model)**." In KCC 2015

Patents

[14] **Byungkook Oh**, Hae-Yoon Koo, and Jinho Choi, "**텍스트 정보량 기반 가변 길이 아이템 토큰화 방법 및 장치**.", KR Patent (Under Review)

[13] **Byungkook Oh**, Jinho Choi, and Hye-Yoon Baek, "**인과성 유도 생성 흐름 네트워크 기반 지식 그래프 검색 방법 및 장치** KR Patent (Under Review)

[12] **Byungkook Oh**, Hye-Yoon Baek, and Hae-Yoon Koo, "**논리 규칙을 내재화한 대규모 언어 모델 기반 정보 추출 방법 및 장치**." KR Patent (Under Review)

[11] **Byungkook Oh** and Dongcheon Lee, "**지식 그래프 기반 질의 응답 시스템의 최적화 방법 및 장치** *(Method and Apparatus for Optimization of a Knowledge Graph-based Question Answering System)*." KR Patent No. 10-2025-0198283, filed December 12, 2025

[10] **Byungkook Oh** and Sangjun Ji, "**하이퍼관계형 지식 그래프 상의 하이퍼관계형 사실에 대한 링크 예측 방법 및 장치** *(Method and Apparatus for Link Prediction of Hyper-Relational Facts on Hyper-Relational Knowledge Graphs)*." KR Patent No. 10-2025-0198284, filed December 12, 2025

[9] **Byungkook Oh** and Hye-Yoon Baek, "**동적 스팬 인식 정보 추출 방법 및 장치** *(Dynamic Span-Aware Information Extraction Method and Apparatus)*." KR Patent No. 10-2025-0162443, filed October 31, 2025

[8] **Byungkook Oh** and Jimyeung Seo, "**멀티모달 데이터 증류 방법, 트리플 기반 뉴스 정제 및 추천 방법** *(Multimodal Data Distillation Method, and Triple-based News Refinement and Recommendation Method)*." KR Patent No. 10-2025-0161770, filed October 31, 2025

[7] **Byungkook Oh** and XXX, "**Electronic Device and Controlling Method of Electronic Device**." *(Explainable User Intent Learning based on Multi-Level Concepts)*, KR Patent, filed October 28, 2022

[6] XXX, XXX, XXX, **Byungkook Oh**, and XXX, "**Server and Control Method Thereof**." *(Pre-training Multi-Modal Geolocation Influence with Contrastive Learning, Dynamic-Positional Graph Neural Network for Next PoI Prediction)* KR Patent, filed February 23, 2022

[5] Kyung-Ho Lee, Sanghak Lee, Seungmin Seo, and **Byungkook Oh**, "**Method and Apparatus for Relation Extraction between Entities**." KR Patent No. 10-2382914, filed June 17, 2020. and issued March 31, 2022

[4] **Byungkook Oh** and Kyong-Ho Lee, "**Method and Apparatus for Completing Zero-Shot Knowledge Graph Using Multi-Hop Neighborhoods**." KR Patent No. 10-2273136, filed October 17, 2019. and issued June 29, 2021

[3] **Byungkook Oh** and Kyong-Ho Lee, "**Method and Apparatus for Reconstructing Sentence Order through Topic-Guided Coherence Modeling**." KR Patent No. 10-2329738, filed October 30, 2019. and issued November 17, 2021

[2] **Byungkook Oh** and Kyong-Ho Lee, "**Method and Apparatus for Completing Knowledge Graph Based on Convolutional Learning Using Multi-Hop Neighborhoods**." KR Patent No. 10-2189811, filed April 5, 2019 and issued December 04, 2020

[1] **Byungkook Oh** and Kyong-Ho Lee, "**Recommender System and Method Considering Correlations among Contexts**." KR Patent No. 10-1997222, filed January 03, 2018. and issued July 01, 2019

Research Projects

제조 도메인 지식그래프 기반 멀티턴 질의응답 기술 개발 *LG전자 생산기술원*
- **Principal Investigator** *May. 2026 - Dec. 2026*

우주항공 제조설비의 멀티모달 이상탐지와 실시간 생성형 AI 보고체계 기반의 AX전환 실증 *경남테크노파크*
- **Principal Investigator** *May. 2026 - Dec. 2026*

그래프기반 위협추적 분석 *LIG넥스원*
- **Principal Investigator** *Oct. 2025 - May. 2027*

추천시스템 모델 설계 자문 및 사례연구 *한화시스템*
- **Principal Investigator** *Sep. 2025 - Aug. 2027*

멀티모달 생성모델의 신뢰성 및 제어가능성 향상을 위한 그래프 구조 기반의 상호작용 *한국연구재단*
- **Principal Investigator** *Sep. 2024 - Aug. 2028*

LLM 기반 추천을 위한 그래프 기반 컨텍스트 추출 에이전트 *실감미디어혁신융합대학사업단*
- **Principal Investigator** *May. 2025 - Nov. 2025*

논리적 패턴을 반영할 수 있는 지식그래프 표현학습 *건국대학교*
- **Principal Investigator** *Apr. 2024 - Mar. 2025*

온톨로지 기반 멀티타입 상호작용 모델링 기술 *Yonsei University (Prof. Kyong-Ho Lee)*
Ontology-based Multi-type Interaction Modeling *Dec. 2022 - Dec. 2023*
- **Project Manager**: R&D Supervision (planning, advice, verification, etc.)

지식그래프 기반 사용자 의도 및 커뮤니티 탐색 기술 *Yonsei University (Prof. Kyong-Ho Lee)*
Knowledge Graph based Intent-Community Detection *Dec. 2021 - Dec. 2022*
- **Project Manager**: R&D Supervision (planning, advice, verification, etc.)

동적으로 진화하는 지식그래프의 정보 품질 개선을 위한 심층 표현학습 *한국연구재단*
Deep Representation Learning for Improving the Information Quality of Dynamically Evolving Knowledge Graphs *Jun. 2019 - Dec. 2020*
- **Project Leader**: General R&D operation (planning, proposal, development, verification, etc.)
- Text-based zero/few-shot inference model development for KGC
- Learning sophisticated KG representations with neighborhood features aggregated by GCNs

액티브 러닝을 활용한 다항 관계 지식 추출 및 구조화 기술 연구 *NC소프트 R&D*
Active Learning based N-ary Relation Extraction and Strucuralization *May. 2019 - Dec. 2020*
- Korean dataset construction
- Relation extraction model development based on Tree-LSTM and Transformer
- Active learning algorithm generate a new semantic type from the extracted relationships

XAI (eXplainable AI) 기반 스마트 에너지 플랫폼 기술 개발 *한전 전력연구원*
Development of XAI-based Technology for Smart Energy Platform *Mar. 2018 - Dec. 2020*
- **Project Leader**: General R&D operation (planning, proposal, development, verification, etc.)
- 가상 마이크로그리드 데이터셋 구축
- 에너지 지식 그래프 설계 및 지식 추론 연구
- 수요예측 딥러닝&확률모형 융합 모델 개발

사물인터넷을 위한 복합 이벤트 처리 기반 개인맞춤 상황인지형 추천 시스템 *한국연구재단*
A Personalized Context-aware Recommendation System *Jun. 2016 - Jun. 2019*
- **Project Leader**: General R&D operation (planning, proposal, development, verification, etc.)
- 지식 추론 모델 개발 (CNN, TF, GCN 등 활용)
- 텐서 분해 기반 추천 시스템 개발

사용자 중심 지식그래프 스키마 확장 기술 *NC소프트 R&D*
User-centeric Knowledge Graph Schema Expansion Technology *Jun. 2018 - Feb. 2019*
- 한국어 데이터셋 구축
- 관계 추출 모델 개발 (Transformer 등 활용)
- 액티브 러닝 알고리즘 개발

통합 주소/식별체계에 기반한 IoT 서비스 플랫폼 *미래창조과학부*
Service Platform for Internet of Things *Apr. 2013 - Apr. 2018*
- 기계 학습 기반 스트림 데이터 처리
- 구조적 데이터 생성 모델 개발

IoV 데이터 저장 및 분석을 위한 자동차 클라우드 서비스 플랫폼 *미래창조과학부*
Cloud-based Service Platform for IoV Data Storage and Analysis *Jun. 2016 - Dec. 2016*
- IoV 플랫폼 기술 검증
- 데이터셋 수집 및 구축

사물 웹에서 빅 데이터의 시맨틱 매쉬업을 위한 클라우드 컴퓨팅 프레임워크 *한국연구재단*
A Cloud Computing Framework for Semantic Mashup of Big Data in the Web of Things *Jun. 2013 - Jun. 2016*
- 웹 기반 데이터 매쉬업 프레임워크 개발
- 데이터 매쉬업 시나리오에 기반한 실제 모형 개발

제로에너지 커뮤니티에 적용한 가능한 시맨틱 서비스 및 기술 연구 *한국전자통신연구원(ETRI)*
Semantic Services and Technologies for Realizing Zero-Energy Community *Aug. 2015 - Dec. 2015*
- 프로슈머 기반 에너지 지식 그래프 설계

개인지식그래프에 기반한 시맨틱 검색 및 추천 *LG전자*
Semantic Search and Service Recommendation Based on Personal Knowledge Graph *Mar. 2015 - Oct. 2015*
- 지식그래프 탐색 알고리즘 개발
- 텐서 분해 기반의 개인맞춤 상황인지형 추천 시스템 개발

Honors & Awards

- BK21 PLUS Yonsei Scholarship in 2016-2, 2018-1, 2018-2, 2019-1, 2019-2, and 2020-1
- Yonsei University Research Scholarship in 2019-1 and 2020-1
- Undergraduate, Software Capstone Design (2014 fall semester), 1st prize
- Undergraduate, 2013 Spring semester GPA honors
- Undergraduate, National S&T(Science & Technology) Scholarship (Full Tuition for 2011)